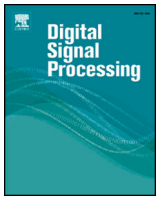
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DeepFake detection with multi-scale convolution and vision transformer



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ABSTRACT

With the help of some modern image generative techniques, it is possible to generate or manipulate image or video contents without introducing any obvious visual artifacts. If these manipulated images/videos are abused, it probably has a huge negative impact on society and individuals. Thus, deepfake detection has attracted considerable attention in recent years. Although the existing methods can achieve good detection performance on high-quality datasets, they are still far from satisfactory for low-quality dataset and cross-dataset evaluation. In this paper, therefore, we propose a new CNN-based method via multi-scale convolution and vision transformer for deepfake detection. In the proposed model, we design a multi-scale module with dilation convolution and depthwise separable convolution to capture more face details and tampering artifacts at different scales. Unlike the traditional classification module, furthermore, we employ a vision transformer to further learn the global information of face features for classification. Extensive experiments demonstrate that in most cases the proposed method achieves better detection results on both high-quality and low-quality datasets compared with related modern methods, and the cross-dataset generalization capabilities of the proposed method are good. In addition, many ablation experiments are provided to verify the rationality of the proposed network.

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1. Introduction

With the rapid development of some modern image generative techniques, especially for generative adversarial networks (GANs), more and more sophisticated deepfake methods have been provided to generate or manipulate realistic faces that people can not distinguish them from real ones. Nowadays, many open source code[1,2] and mobile applications (e.g., FaceApp[3] and Zao[4]) are available so that we can easily and effectively create manipulated face images and videos without any specialized knowledge [41]. If these face faces are used for nefarious purposes, for things like revenge, porn or fraud, which would have a huge negative impact on society and individuals. Thus, it is important to develop effective methods for deepfake detection.

Up to now, many deepfake detection methods based on convolutional neural network (CNN), such as [1,20,29,31,22,3,26,27,14,37,34], have been proposed. Typically, most existing methods take deepfake detection as a binary classification problem, and they often employ a convolutional neural network to extract features of the suspect face images and get the final classification by one or several layers of full connection. The modern methods can achieve very good detection performances on those high-quality images (or image frames within a video). However, images or videos probably are undergone lossy compression and/or other post-processing in real applications so that the quality would decline. As illustrated in Fig. 1, many tampering artifacts introduced by deepfake would be reduced and become blurring for the low-quality versions. In this case, the corresponding performance of deepfake detectors would drop significantly. Up to now, the detection performances of existing methods are still far from satisfactory for those low-quality images. In addition, their generalization performances need further enhancement too.

In this paper, we propose a new CNN-based method via multi-scale convolution and vision transformer for deepfake detection. Specifically, we employ a dual-subnet structure to catch high-level face semantic information and employ a vision transformer module for the final classification in the proposed model. The dual-subnet includes a multi-scale module branch and a MBConv

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[1] https://github.com/deepfakes/faceswap.
[2] https://github.com/MarekKowalski/FaceSwap.
[3] https://apps.apple.com/gb/app/faceapp-ai-face-editor/id1180884341.
[4] https://apps.apple.com/cn/app/id1465199127.

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(a) Example #1 (HQ) (b) Example #2 (HQ)

(c) Example #1 (LQ) (d) Example #2 (LQ)

Fig. 1. Two fake face examples with different quality from FaceForensics++ [35]. Images in the first row are high-quality (HQ), while those in the second row are the corresponding low-quality (LQ) versions after lossy compression.

blocks module branch. In the multi-scale module branch, we first set 4 dilation convolutions with different dilation ratios to capture the face details at different scales, and then we concatenate the extracted face detail information at different scales, and refine the concatenated features with depthwise separable convolutions for the model to fully learn the local discontinuities of manipulated images. In the MBConv blocks module branch, we learn high-dimensional face semantic information by stacking multiple MBConv Blocks. In the final classification, we introduce a vision transformer module to replace global average pooling (GAP) and fully connection layers (FC) used in most existing methods. Extensive comparative experiments show that the superiority of the proposed method. In summary, we propose a dual-subnet structure and combines vision transformer for deepfake detection and the main contributions of this paper are as follows:

- We propose a dual-subnet structure which combines the multi-scale module and the MBConv block module for deepfake detection, and employ the vision transformer for final classification.
- We carefully design and analyze each important module in the proposed model. For instance, we propose a multi-scale module and show that the multi-scale module can effectively capture the face details at different scales and enhance the detection performance of the proposed model.
- Extensive comparative results show that the proposed method outperforms the most modern methods and achieves advanced detection performance on both high-quality and low-quality databases, and the generalization ability of the proposed method is also good.

The rest of this paper is arranged as follows. Section 2 describes the relative work. Section 3 describes the proposed method in details. Section 4 shows comparative results and discussions. Finally, the concluding remarks of this paper and future works are given in Section 5.

Table 1
Six deepfake databases commonly used in recent literatures. LQ denotes low-quality, while HQ denotes high-quality.

Dataset	Real	Fake	Year
UADFV [23]	49	49	2018
Deepfake-TIMIT [18]	320	LQ:320 HQ:320	2018
Celeb-DF [25]	890	5639	2019
DFDC Preview [8]	1131	4119	2019
DFDC Full [9]	20,000	100,000	2020
FaceForensics++ [35]	RAW:1000 LQ:1000 HQ:1000	RAW:4000 LQ:4000 HQ:4000	2019
WildDeepfake [48]	3805	3509	2020

2. Relative work

In this section, we will give a brief overview of deepfake technique, deepfake datasets, and deepfake detection methods that used in recent literatures.

2.1. DeepFake technique

Typically, the existing deepfake techniques can be divided into 4 categories [41]: entire face synthesis, identity swap, attribute manipulation, and expression swap. Entire face synthesis aims to generate entire non-existent face images, the typical methods include the StyleGAN [16] and PGGAN [15]; identity swap tries to swap the face of one person (source face) in videos with the face of another person (target face), such as FSGAN [32] and FaceShifter [21]; attribute manipulation aims to modify some attributes (e.g., hair or skin) of the target face, such as StarGAN [5] and AttGAN [12]; expression swap tries to change the facial expression (e.g., excitement and frustration) of the target face, such as NeuralTextures [39] and Face2Face [40].

Among the four categories, identity swap is currently recognized as the most harmful and widely studied type of face tampering [41]. For tampering a video, this technique usually includes three steps: it firstly divides the video into frames and locates the faces in each frame; and then it swaps the face of a person (i.e., source face) in a frame with the face of another person (i.e., target face) through some deep learning methods. Finally, it applies some image fusion techniques to stitch face images to reduce tampering artifact and synthesize a fake video. Up to now, many effective identity swap methods have been proposed. For instance, Korshunova et al. [19] proposed the first face swap method to transform the identity. They use convolution neural networks to capture the appearance of the target images and propose a new loss function to produce highly realistic results; Olszewski et al. [33] propose a method to realistically puppeteer and animate a face from RGB images using a source video sequence; Li et al. [21] try to generate the swapped face with high-quality by exploiting and integrating the target attributes thoroughly and adaptively, and propose a novel two-stage face swapping method, called FaceShiter.

2.2. DeepFake datasets

Up to now, there are some public deepfake datasets available online. As illustrated in Table 1, we will describe six typical datasets that are commonly used for comparative studies in recent literatures in the following.

- UADFV [23] includes 49 real videos from Youtube, which are then used to generate 49 fake videos. These videos are about 11.14 seconds, and each video represents one identity and has the resolution of 294×500.

- Deepfake-TIMIT [18] includes 320 real videos and 640 fake videos. Among fake videos, there are 320 high-quality videos and 320 low-quality videos separately.
- Celeb-DF [25] includes 890 real videos and 5,639 fake videos. The videos in this dataset have better visual quality and higher resolution.
- DFDC [8] includes two different versions: preview version and full version. The preview version includes 1131 real videos from 66 paid actors and 4119 fake videos generated by 2 different unknown approaches, while the full version includes more than 20,000 real videos and more than 100,000 fake videos generated by 8 different unknown approaches.
- FaceFornesics++ [35] includes 1,000 original videos from YouTube, and then generates the corresponding fake videos with four deepfake techniques, called DeepFakes,[5] FaceSwap, Face2Face [40], and NeuralTexture [39] separately. Note that three different quality versions are considered in this dataset: RAW denotes the original videos without any post-processing; HQ denotes the videos compressed with a fixed rate quantization parameter of 23; LQ denotes the videos compressed with the fixed rate quantization parameter of 40. For each quality version, therefore, there are 1,000 real videos and 4,000 manipulated videos.
- WildDeepfake [48] includes 3,805 real videos and 3,509 fake videos that are downloaded from the Internet, meaning that these videos come from various sources. What is more, they probably have been compressed more than one time. Thus, the quality of videos in this dataset is relatively poor.

Note that the numbers of videos in UADFV and Deepfake-TIMIT are relatively fewer, and the modern deepfake detection methods can achieve better performances on the two datasets. In this paper, therefore, we employ the other 4 datasets (i.e., Celeb-DF, DFDC, FaceFornesics++, and WildDeepfake) that are more difficult to detect in our experiments.

2.3. Deepfake detection method

In the past several years, many CNN-based methods, such as [1,20,29,31,22,3,26,27,14,37,24,34], have been proposed for deekfake detection. For example, Afchar et al. [1] focused on the mesoscopic properties of images, and proposed a MesoNet to detect online videos; Hguyen et al. [31] first introduced the capsule networks to detect forged images and videos; Li et al. [20] analyzed the differences among the color components within the generated images for classification; Rössler et al. [35] first introduced the Xception model into deepfake detection, and show that the detection performances of Xception can outperform most related methods at that time; Li et al. [24] proposed a new method that can effectively distinguish manipulated videos from real videos without using fake images as negative training; Qian et al. [34] employed two different but complementary frequency-aware clues to describe subtle forgery artifacts and proposed a new method (F3-Net) for deepfake detection; Zhao et al. [45] formulated deepfake detection as a fine-grained classification problem and proposed a new multi-attentional deepfake detection network; Li et al. [22] proposed a new image representation called Face-X-Ray to show the blending boundary for a forged image and the absence of blending for a real image, and achieved a high detection accuracy and generalization performance in high-quality videos; Castillo et al. [3] proposed two different methods of initializing convolutional neural networks for image manipulation detection; Shang et al. [37] proposed a Pixel-Region Relation Network (PRRNet) to capture pixel-wise and region-wise relations respectively for deepfake detection; Jia et al. [14] proposed an inconsistency-aware wavelet dual-branch network to detect inter-image and intra-image inconsistencies in deepfake; Liu et al. [26] employed spatial image and phase spectrum to capture the up-sampling artifacts of deepfake to improve the transferability, and achieved the very good results on multi-class classification on FaceFornesics++ dataset.

Note that some modern methods, such as Dual-Branch [14] and PRRnet [37], can achieve very good performances for those high-quality images (e.g., on Celeb-DF [25] and high-quality versions of FaceForensics++ [35]). However, the detection performances are still from satisfactory for low-quality images. In addition, the generalization ability of existing methods needs further improvement.

3. Proposed method

As illustrated in Fig. 2, The network architecture includes four different modules, that is, pre-processing module, multi-scale module, MBConv block module, and vision transformer module. Note that here we select the EfficientNet-b4 [38] as the backbone of the proposed method (see the pre-processing module and MBConv block module) since the EfficientNet-b4 can provide a good trade-off in dimensionality, computing time, and classification performance on deepfake detection. Thus EfficientNet-b4 usually serves as a backbone or baseline in many existing literatures. In the following, we will describe the four modules in detail.

3.1. Pre-processing module

The pre-processing module includes a 3×3 convolution, 2 MBConv1 Blocks and 8 MBConv6 Blocks, which is defined in EfficientNet-B4 [38]. As reported in [38], the EfficitentNet-b4 is one version of the EfficientNet, which consists of 2 classical convolutions, 2 MBConv1 blocks, 30 MBConv6 blocks and a fully connected layer. Besides, the EfficientNet-b4 has good image classification performance with 19 millions of parameters and 4.2 billions of FLOPs and reaches the 83.8% top-1 accuracy on the ImageNet database. The network architecture of MBConv1 block and MBConv6 block which are defined in EfficientNet-b4 are as illustrated in Fig. 3. Note that the MBConv1 block consists of Depthwise convolution, Batch Normalization (BN), Squeeze-and-Excitation module (SE) [13] and Pointwise convolution & BN. Compared with the MBConv1 block, the MBConv6 block adds a Pointwise convolution and a BN layer.

For a face image, the pre-processing module aims to produce a preliminary feature map $f_p \in \mathbb{R}^{H \times W \times C}$, where (H, W) is the feature resolution and C is the number of the feature channel. Then f_p is then fed into a dual-subnet in the proposed model.

3.2. Dual-subnet: multi-scale module & MBConv block module

In the proposed network architecture, we employ a dual-subnet structure, that is, a multi-scale module and a MBConv block module, to deal with the feature map f_p which is obtained in the pre-processing module. The multi-scale module aims to learn multi-scale face information, while the MBConv block module aims to learn high-dimensional face semantic information.

3.2.1. Multi-scale module

Most existing CNN-based methods for deepfake detection, such as [1,29,31], usually stack single-scale convolutional layers to extract high-dimensional face semantic information. In the proposed method, however, we propose a multi-scale module with different dilation convolutions and depthwise separable convolutions in

[5] https://github.com/deepfakes/faceswap.

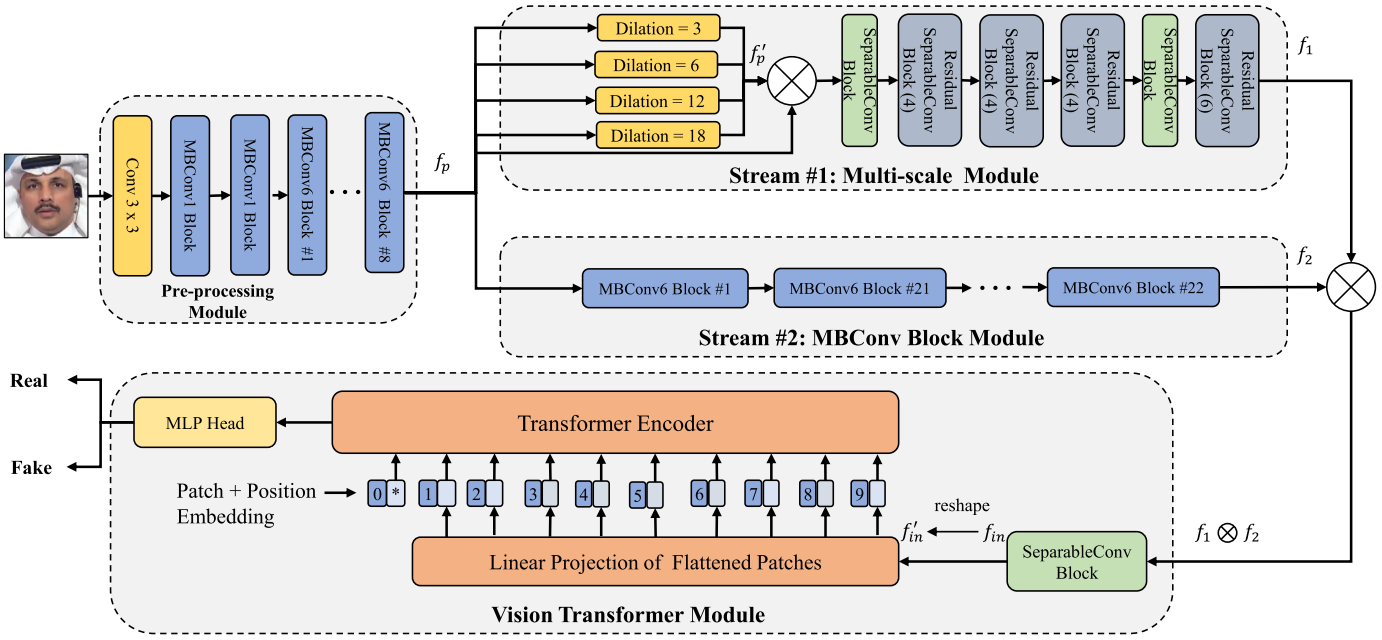


Fig. 2. The network architecture of the proposed method. It includes four modules: pre-processing module, multi-scale module, MBConv block module and vision transformer module. The symbol Dilation = N denotes the dilation rate is N. The symbol $\otimes$ denotes the operation of concatenation.

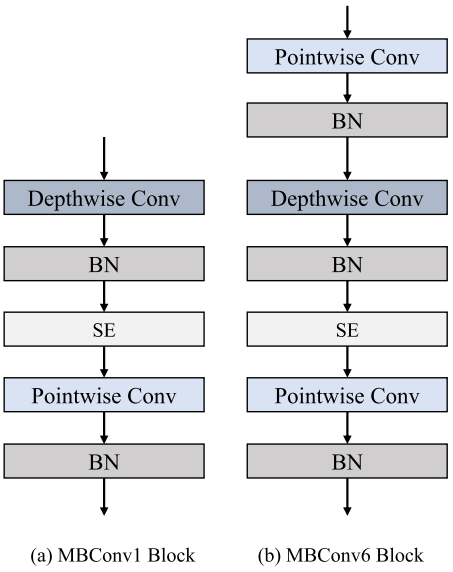


Fig. 3. The network architecture of the MBConv1 Block and MBConv6 Block used in the pre-processing module and MBConv block module.

order to capture more face details and tampering artifacts at different scales. The dilation convolution and depthwise separable convolution used in this module are described as follows:

- **Dilation convolution:** Firstly, we use dilation convolutions with four different dilation rates (i.e., 3, 6, 12, and 18) to process the input feature map in order to extract multi-scale face information of the feature map f_p. In this way, we can obtain four different scale face feature maps which have the same resolution and channel as f_p. Finally, we concatenate the four different scale face feature maps and the original one (i.e., f_p) to get a multi-scale face feature map $f'_p \in \mathbb{R}^{H \times W \times 5C}$
- **Depthwise separable convolution:** And then, we employ 2 SeparableConv Blocks and 4 Residual SeparableConv Blocks to learn channel correlation and spatial correlation separately for

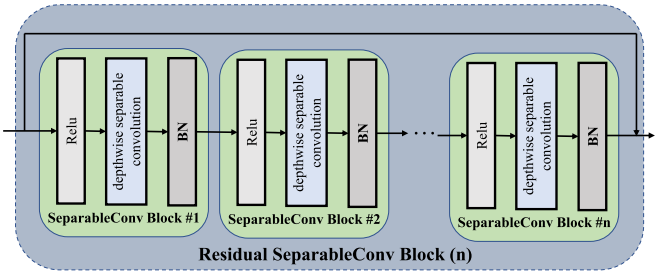


Fig. 4. The network architecture of "Residual SeparableConv block (n)" used in the multi-scale module, where n denotes the number of the separableConv block used in this architecture.

the feature map f'_p. Specifically, each separableConv block contains a relu non-linearity, a depthwise separable convolution, and a batch normalization. The first three residual separableConv blocks include 4 separableConv blocks (i.e., Residual SparableConv Block (4) as illustrated in Fig. 4), while the last residual separableConv block includes 6 separableConv block (i.e., Residual SparableConv Block (6)), and each residual separableConv block has a linear residual connection. In this way, we can refine the f'_p, learn more multi-scale face details information, and capture manipulated artifacts at different scales effectively. In summary, the proposed depthwise separable convolution block can enhance the representation ability of the model and get the richer face feature of manipulated images.

Finally, we can get a high-dimensional multi-scale feature map, denoted as f_1, after dilation convolution and depthwise separable convolution as described about.

3.2.2. MBConv block module

The proposed MBConv block module follows the same setting that are used in the EfficientNet-B4 [38]. Specifically, we stack 22 MBConv6 Blocks together to obtain a high-dimensional face semantic feature map f_2 for face forgery detection. Note that the MBConv Block architecture [36] (both MBConv1 block and MB-

Conv6 block that are illustrated in Fig. 3) is based on an inverted residual structure. Compared with the traditional residual models, the MBConv blocks use lightweight depthwise convolutions to filter features in the intermediate expansion layer. In this way, we can effectively reduce the model parameters while ensuring the accuracy of the model based on our experiments.

Finally, we concatenate the resulting feature maps (i.e., f_1 in Stream #1 and f_2 in Stream #2) in the dual-subnet for subsequent processing. Note that here f_1 and f_2 have the same resolution and channel.

3.3. Vision transformer module

Most existing methods, including the EfficientNet-b4 and methods [1,31,27,14], usually feed the concatenated feature map (i.e., $f_1 \otimes f_2$) into a global average pooling and full connection for the final classification. Inspired by [7], we employ a vision transformer module for classification. Firstly, we use a separableConv block to increase the dimension of $f_1 \otimes f_2$ and output f_{in}. Since the standard transformer receives a 1D sequence of token embedding as input, we reshape f_{in} into f'_{in} as it did in [10]. Then, we flatten f'_{in} and map the flattened f'_{in} into D dimensions by trainable liner projection, where the value of D is the size of the constant hidden vector which is used in the all layers of transformer. And then the position embeddings are added to the patch embedding to retain the positional information. The result of embedding is used as the input of the transformer encoder. The transformer encoder is composed of alternating layers of multi-head self-attention and multilayer perceptron (MLP) blocks. In this way, the transformer can not only integrate the information of the entire feature maps via self-attention, but also understand the image structure through the learnable position embedding. Finally, we can get the final classification result through an MLP.

4. Experiments

Four popular datasets, including Celeb-DF [25], DFDC [8], FaceForensics++ (denoted as FF++) [35], and WildDeepfake [48] are considered in our experiments. First of all, we use the MTCNN [44] to detect and extract faces for each frame and enlarge the face region by a factor of 1.3 around the center of the detected face, and then we resize the face images into 320×320 as the input of the proposed method. For a fair comparison, all experimental results are evaluated on the same test dataset for all compared methods. Like most related methods, we directly use the reported performances of existing methods for comparative study.

In training stage, we employ the cross-entropy and Adam optimizer with the batch size 48. We train the proposed networks in total of 20 epochs and reduce the learning rate to 1/10 of the original every 10 epochs. We set the initial learning rate to 0.0001 and select the model with the best accuracy as the final model.[6]

4.1. Comparative results with related methods

To show the effectiveness of the proposed method, we compare the proposed method with some modern methods on four different datasets respectively. Note that the source codes of most existing methods are not available, and most existing methods do not provide detection results on the four datasets simultaneously. Like related methods, therefore, the compared methods are different for different datasets. However, the comparison is fair since all the detection performances are evaluated on exactly the same partition (i.e., training, verification and test set) in each dataset.

[6] The source codes are now available on https://github.com/mouse1998mouse/muti-scale-vit so that the readers can understand the proposed model clearly and repeat our experimental results easily.

Table 2
The AUC (%) evaluated on the Celeb-DF dataset for different methods. Note that in the following tables, those highlighted values with an asterisk "*" denote the best results in the corresponding cases.

Method	AUC
Xception [6]	97.60
Multi-task [30]	90.50
Capsule [31]	93.20
DSW-FPA [24]	94.80
DCViT [42]	97.20
PRRnet [37]	**99.80** *
Dual-Branch [14]	99.50
Zhou et al. [47]	99.78
Proposed Method	**99.80** *

Table 3
The AUC (%) evaluated on the DFDC dataset for different methods.

Method	AUC
Meso-4 [1]	82.20
Meso-Inception [1]	81.76
DSLRFN [17]	84.48
DSP-FWA [24]	85.26
Xception [6]	85.89
Proposed Method	**88.47** *

4.1.1. Experiments on Celeb-DF

In this section, we use the Celeb-DF for evaluation. As described in Section 2.2, the videos in Celeb-DF have better visual quality and higher resolution. Following the original data division, we use the designated 518 videos for testing. In the remaining videos, we randomly select 20% videos as the validation set. In the training and validation stage, we extract 50 image frames for each manipulated video and original video. Like related methods, AUC (area under the receiver operating characteristic curve) is used as evaluation metric since the class in this dataset is imbalance.

The comparative results are shown in Table 2. From the Table 2, we observe that all AUC of modern methods are higher than 90%, which means that this database is relatively easier to detect due to its high-quality. Both the proposed method and the PRRNet [37] can achieve the current best performance with the AUC as high as 99.80%.

4.1.2. Experiments on DFDC

Like most existing methods, such as [17,24], we employ the preview version of DFDC for evaluation in the section. As described in Section 2.2, this version contains 1131 real videos and 4105 fake videos. In our experiment, we follow the same data division and randomly select 20% training videos as the validation set. In the training and validation stage, we extract 50 image frames for each manipulated video and original video. Similarly, AUC on the test dataset is considered due to the imbalance of data categories.

The comparative results are shown in Table 3. From Table 3, we observe that the proposed method achieves the best performance in AUC of 88.47%. Compared with the Xception [6], the proposed method achieves 2.58% improvement in AUC. From Table 3 and Table 2, we observe that compared with Celeb-DF, the DFDC dataset is much more difficult to detect since it employs a variety of unknown algorithms to generate manipulated videos.

Table 4
The ACC (%) and AUC (%) evaluated on the FF++(LQ) dataset for different methods.

Method	ACC	AUC
MesoNet [1]	70.47	-
Face X-ray [22]	-	61.60
Xception [6]	86.86	89.30
F3-Net [34]	90.43	93.30
Two Branch [27]	-	86.59
EfficientNet-b4 [38]	86.67	88.20
Multi-attention [45]	88.69	90.40
Dual-Branch [14]	88.96	92.97
Spatial-Phase [26]	81.57	82.82
CFFExtractor [43]	-	90.35
PRRnet [37]	86.13	-
FRLM [28]	90.71	93.69
Patch diffusion [2]	86.11	89.29
Proposed Method	**90.74** *	**94.86** *

Table 5
The ACC (%) evaluated on the WildDeepfake dataset for different methods.

Method	ACC
ResNetv2-50 [11]	63.99
MesoNet [1]	64.74
Meso-inception [1]	66.03
Xception [6]	76.27
ADDNet-2D [48]	76.25
Zhou et al. [47]	79.87
Proposed Method	**82.63** *

4.1.3. Experiments on FF++ (LQ)

In this section, we use the low quality (LQ) videos in FaceForensics++ (denoted as FF++ (LQ)) for evaluation. This dataset contains 1,000 original videos collected from YouTube, and each original video has been manipulated with four different deepfake techniques separately. Like existing methods, we divide the original video clips (as well as the corresponding fake versions) into training, validation and test parts, which contain 720, 140, and 140 original videos respectively. In training stage, to alleviate the class imbalance problem, we extract 200 frames from each original video and extract 50 frames ($= 200 \times \frac{1}{4}$) from the 4 corresponding fake video. Like related methods, both ACC (accuracy) and AUC are used as evaluation metrics in this dataset.

The comparative results are shown in Table 4. From Table 4, we observe that the proposed method always achieves the best results with regard to both ACC and AUC. Note that the ACC and AUC of most existing methods are less than 90%, while the proposed gets 90.74% ACC and 94.86% AUC. Compared with the current best method (i.e., FRLM [28]), the proposed method can still obtain 0.03% improvement on ACC and 1.17% improvement on AUC.

4.1.4. Experiments on WildDeepfake

In this section, we use the WildDeepfake dataset for evaluation. Note that the WildDeepfake is a challenging real-world dataset for deepfake detection. Unlike other datasets, both real and fake videos in WildDeepfake are collected from the Internet. Therefore, these videos are probably from different sources and/or compressed several times, which makes the detection very difficult.

The comparative results are shown in Table 5. From Table 5, we observe that the detection accuracies of existing methods are less than 80%. However, the proposed method can achieve state-of-the-art result with the detection accuracy as high as 82.63%. Compared with the current best method (i.e., method [47]) on this database, we can still obtain 2.76% improvements.

4.2. Cross-dataset evaluation

In this section, we evaluate the generalization performance of the proposed model. In the following, two different cross-dataset evaluations are included in our experiments.

4.2.1. Evaluation on unknown deepfake techniques within FF++

As described in section 2.2, there are 1,000 original real videos in FF++. For each original video, there are four different fake videos generated by four manipulated techniques, including DeepFake (denoted as DF), Face2Face (i.e., F2F), FaceSwap (i.e., FS), and NeuralTextures (i.e., NT). In this section, we firstly train a classifier on the images generated by one manipulated technique, and then use the resulting classifier on other images generated by other three manipulated techniques. For a fair comparison, all the experimental results are evaluated on raw quality images without any post-processing as it did in [4]. The comparative results are shown in Table 6. From Table 6, we have three following observations:

- If the training data and test data are generated by the same manipulated technique, the performances of the proposed method are all over 99.25%, as shown in the diagonal results of the table. Similar good results can be obtained for the other two detection methods (i.e., Xception and Res-Layer1), which means that the tampering artifacts introduced by the four manipulated techniques are relatively easy to detect.
- If the training dataset does not match the test dataset, the corresponding performances would decline in different ranges for the three detection methods. Taking DF as training dataset for instance, our detection accuracies would drop from 99.97% to 90.32%, 44.73% and 94.53% when evaluating the F2F, FS and NT separately, which means that the inherent statistical properties of the fake images generated by different manipulated techniques are different.
- Overall, the proposed method outperforms two other detection methods (i.e., Xception and Res-Layer1) in most cases. The detection improvements are significant in some cases. Taking FS as training dataset for instance, our detection accuracy on NT is as high as 96.24%, while they are less than 70% for the other two methods, which means that the cross-dataset generalization capabilities of the proposed method are better.

Note that when we use F2F as our training dataset, the results are relatively poor for FS. The main reason is that statistical differences between F2F and FS are relatively huge.

4.2.2. Evaluation on unknown dataset

As it did in most related methods (such as [34,45]), in this section, we firstly train the proposed model on the FF++ (HQ) dataset, and then use the resulting classifier to evaluate those images from the Celeb-DF dataset. The comparative results are shown in Table 7. From Table 7, we observe that the detection results are lower than 68% for all existing methods, while the proposed method achieves result as high as 71.41%. Compared with the current best method (i.e., multi-attention [45]), we can still achieve 3.97% improvements, which means that the proposed method has much better generalization ability compared with existing ones.

4.3. Ablation study

In order to verify the rationality of the proposed network, some ablation studies are conducted in this section. For simplification, just the WildDeepfake dataset is used for performance evaluation. WildDeepfake is a more challenging dataset, and using this dataset for ablation experiments can amplify the variance of each module in the proposed method.

Table 6
The AUC(%) evaluated on different manipulated methods. Those highlighted values with an asterisk "*" denote the best results in the corresponding cases. The results of other methods are from related papers.

training set	method	DF	F2F	FS	NT
DF	Xception [6]	98.38	75.05	49.13	80.38
	Res-Layer1 [4]	98.97	71.74	**57.15** *	74.99
	Proposed Method	**99.97** *	**90.32** *	44.73	**94.53** *
F2F	Xception [6]	**87.56** *	**99.53** *	**65.23** *	65.90
	Res-Layer1 [4]	84.39	97.66	60.53	**79.72** *
	Proposed Method	86.13	99.26	50.16	74.53
FS	Xception [6]	70.12	61.70	99.36	68.71
	Res-Layer1 [4]	59.49	62.00	97.13	52.56
	Proposed Method	**89.50** *	**86.74** *	**99.97** *	**96.24** *
NT	Xception [6]	93.09	84.82	47.98	**99.50** *
	Res-Layer1 [4]	70.32	65.04	53.37	86.93
	Proposed Method	**99.41** *	**89.57** *	**66.69** *	99.47

Table 7
Cross-dataset evaluation on Celeb-DF(AUC(%)) by training FF++(HQ). Some results of other methods are cited directly from [45]. Note that in all following tables, those highlighted values with an asterisk "*" denote the best results in the corresponding cases.

Method	AUC
Two-stream [46]	53.80
Meso-Inception [1]	53.60
FWA [24]	56.90
DSP-FWA [24]	64.60
Multi-task [30]	54.30
Capsule [31]	57.50
F3-Net [34]	65.17
Xception [6]	65.30
Multi-attention [45]	67.44
Proposed Method	**71.41** *

Table 8
Ablation results on WildDeepfake dataset. Note that * indicates that we adjusted its learning rate to 1/10 of the original.

Variants	ACC
w/o Vision Transformer module	80.17
w/o Multi-scale Block module	79.53
w/o MBConv Block module	78.13
Proposed Method	**82.63** *

Table 9
Ablation results on the proposed pre-processing module.

number	ACC
7	78.83
9	81.00
8 (the proposed method)	82.63

Table 10
Detection accuracy (%) on WildDeepfake dataset with different quantity dilation convolutions.

Number	Dilation Rate	WildDeepfake
0	-	76.29
3	3, 6, 12	77.10
3	6, 12, 18	78.62
4	3, 6, 9, 12	80.09
4 (Proposed)	3, 6, 12, 18	**82.63** *
5	3, 6, 9, 12, 18	80.85

4.3.1. Analysis on the proposed modules

In this section, we show the detection accuracies when an important module in the proposed model is removed or replaced, including vision transformer module, MBConv blocks module and Multi-scale block module separately. The comparative results are shown in Table 8. From Table 8, we can obtain three following observations:

- First of all, the proposed method can achieve the detection accuracy as high as 82.63%, which are higher than three variants of the proposed method as well as some modern methods (refer to Table 5);
- For the final classification, we replace the vision transformer module as the typical classification module, that is the global average pooling and full connection. The corresponding detection accuracy will drop from 82.63% to 80.17%, which means that capturing the global information via vision transformer are helpful to detect deepfake images;
- The proposed network employs a dual-subnet architecture. If any one of streams in the proposed network is removed, i.e., Stream #1: Multi-scale Block module or Stream #2: the MBConv Block module, the detection accuracies will drop to 79.53% or 78.13% respectively, which means that combining the multi-scale face information (i.e., f_1) and high-dimensional face semantic information (i.e., f_2) is useful to enhance the detection performance.

4.3.2. Analysis on the pre-processing module

In the proposed pre-processing module, we use a 3×3 convolution, 2 MBConv1 blocks and eight MBConv6 block to extract the feature of the manipulated images. In order to ensure the rationality of using 8 MBConv6 blocks, we preform ablation experiments. We reduce the 8 MBConv6 blocks in the pre-processing module to 7 and increase to 9. The experimental results are shown in the Table 9.

From the Table 9, we observe that the number of MBCov6 would affect the final detection performance of the proposed model. Using 8 MBCov6 can achieve the best results (i.e., 82.63%). When adding/reducing one MBCov6 in the proposed model, the corresponding detection accuracy will drop around 1.63% - 3.8%.

4.3.3. Analysis on the dilation convolution in the multi-scale module

In the proposed multi-scale module, we respectively use four dilation convolutions with the dilation rates of 3, 6, 12 and 18 to obtain the multi-scale face information of the feature map f_p, which is generated in pre-processing module. Some comparative experiments evaluation on different dilation convolutions is shown in Table 10. From Table 10, we observe that two following observations:

- First of all, the proposed method always works the best compared with other five variants. If all dilation convolutions are removed (refer to the first row in Table 10) in the proposed



Fig. 5. Illustration of the output feature map (i.e., f_p) of the pre-processing module, and its four modified versions after dilation convolution with 4 different dilation rates (i.e., Dilation =3, Dilation =6,Dilation =12, and Dilation =18).

network, the detection performance would become the poorest, which means that the dilation convolution is very necessary in our model.

- On average, the detection accuracy of the proposed model will drop around 1.78% - 5.53% if adding a dilation convolution with a dilation rate of 12, removing a dilation convolution with a dilation rate of 18 or 6, or changing a dilation rate from 18 to 9, which means that the parameters (i.e., number and dilation rate) of the dilation convolution will also significantly improve the detection performance of the proposed model.

Fig. 5 illustrates some output feature maps (i.e., f_p) of the pre-processing module and the four corresponding modified versions after the dilation convolution with a dilation rate of 3, 6, 12, and 18 separately. Compared with normal convolution, dilation convolution can increase the receptive field and get multi-scale information about the image. From Fig. 5, we observe that the feature maps after different dilation convolutions can provide more details of the input face image. It is well known that most manipulated face images generated by the modern deepfake techniques are visually imperceptible, those tampering artifacts usually reflected in image details. Therefore, it is expected that the dilation convolution used in the proposed model can enrich the face details at different scales and thus enhance the detection performance.

5. Conclusion

In this paper, we propose a new CNN-based method combining with multi-scale feature and vision transformer for deepfake detection, and show that the proposed method can achieve the advanced results evaluated on four popular datasets. The proposed method achieves the state-of-the-art result both on FF++ and Wild-Deepfake. Taking Wilddeepfake for instance, the proposed method can obtain 2.76% improvement on ACC compared with the current best method [47]. In addition, we also show that the generalization ability of the proposed model is good by cross-database evaluation. Furthermore, we provide many ablation studies to verify the rationality of the proposed network.

In the proposed method, face images in the spatial domain are considered. In future, the multi-scale information will be extracted in both the spatial and frequency (such as DCT and/or wavelet) domains for further improving detection performances. Combined with image segmentation technology, furthermore, some content adaptive schemes will be integrated in the proposed network so that the features can be focused on different facial parts, such as eyes, nose, and mouth, which probably have more tampering artifacts.

CRediT authorship contribution statement

Hao Lin: Formal analysis, Methodology, Writing – original draft. **Wenmin Huang:** Writing – review & editing. **Weiqi Luo:** Methodology, Writing – review & editing. **Wei Lu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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